

SAPA Deployment & Integrasi (VPS Ubuntu + Nginx + Kubernetes + MQTT + Edge Server + ESP32)

Dokumen ini menjelaskan instalasi end-to-end SAPA setelah penambahan fitur:

- Live camera dashboard mengambil frame dari '/edge' (kamera browser di-Dashboard sudah dinonaktifkan).
- Tombol ****Open / Close Gate**** manual untuk ****manager + admin**** di tab System, lengkap dengan badge status MQTT (online / offline) dari perangkat IoT.
- Edge Server (Python) menjalankan model ****AI Face Recognition**** secara lokal, menarik gambar referensi dari VPS.
- ESP32 (servo + buzzer + PIR) terhubung ke MQTT ('sapa/gate', 'sapa/pir', 'sapa/device/heartbeat').
- Login admin/manager (sukses + gagal) tersimpan di MongoDB VPS dengan format **** (Username, Status, Timestamp)****, dan ditampilkan real-time di dashboard.
- Tombol ****View Faces**** di tabel Employees untuk melihat foto wajah yang tersimpan.

1) Arsitektur

```
'''
????????? HTTPS ??????????????????
? Browser????????????? Nginx (VPS) ?
??????????? ? ? / ? static frontend (React build)
? ? /api/* ? reverse proxy ke FastAPI :8080
????????????????????
?
????????????????????????????????????
? ? ?
???????????????????????? ????
? FastAPI ? ? MongoDB (k8s)? ? Mosquitto ?
? - /edge/frame ? ? attendance + ? ? (k8s atau ?
? - /edge/face- ? ? audit logs ? ? host) ?
? match ? ?????????????????? ??????????????????
? - /gate/* ? ?MQTT pub/sub
? - /iot/status ? ?
? - /audit/logins? ?
????????????????????????????
? HTTPS ?
? ?
???????????????????????????? MQTT (LAN) ?????????????????????
? Edge Server ????????????????????? ESP32 (gate) ?
? - camera capture ? ? servo + buzzer + ?
? - face_recognition (AI) ? PIR ?
? - sync embeddings dari VPS ? ?
???????????????????????????? ?????????????????????
'''
```

Topik MQTT yang dipakai:

Topic	Arah	Contoh payload
'sapa/gate'	Backend ? ESP32	{ "action": "open", "employee_id": "123456", "ts": "..." }
'sapa/attendance'	ESP32/Edge ? Backend	{ "employee_id": "123456", "is_valid": true, "direction": "in" }
'sapa/device/heartbeat'	ESP32/Edge ? Backend	{ "online": true, "source": "esp32" }
'sapa/pir'	ESP32 ? Backend	{ "motion": true } -> backend kirim 'close' ke gate

2) Prasyarat VPS

- Ubuntu 22.04 LTS, public IP, domain 'sapa.example.com' ke A record.
- Akses SSH (key-based), user 'deploy'.
- Software: Nginx + Certbot, Python 3.11, Node 18+, k3s (Kubernetes lite).

```
'''bash
sudo apt update && sudo apt -y upgrade
sudo apt install -y nginx ufw fail2ban git build-essential cmake \
python3.11 python3.11-venv python3-pip ca-certificates curl
sudo ufw allow OpenSSH && sudo ufw allow 80/tcp && sudo ufw allow 443/tcp && sudo ufw enable
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
```

```
sudo apt install -y nodejs
sudo apt install -y certbot python3-certbot-nginx
'''
```

3) Install k3s + namespace SAPA

```
'''bash
curl -sL https://get.k3s.io | sh -s - --disable traefik
sudo kubectl get nodes
kubectl apply -f k8s/namespace.yaml
'''
```

4) Deploy MongoDB + Mosquitto di k3s

```
'''bash
kubectl apply -f k8s/mongo-deployment.yaml
kubectl apply -f k8s/mqtt-deployment.yaml      # untuk dev / awal
# kubectl apply -f k8s/mosquitto-production.yaml # production: ACL+auth+PVC
# kubectl apply -f k8s/mosquitto-nodeport.yaml  # bila ESP32 LAN
'''
```

> Production: gunakan 'mosquitto-production.yaml' + 'mosquitto-nodeport.yaml'. Buat secret 'sapa-mosquitto-auth' dari file 'mosquitto_passwd', dan UFW allow hanya subnet LAN ke port '31883'.

5) Deploy Backend di VPS (systemd) + env baru

```
'''bash
sudo mkdir -p /opt/sapa && sudo chown -R deploy:deploy /opt
cd /opt && git clone <REPO_URL> sapa
cd sapa/backend && python3.11 -m venv venv
./venv/bin/pip install -r requirements.txt
'''
```

Buat '/opt/sapa/backend/.env':

```
'''
DATABASE_URL=postgresql://user:password@<POSTGRES_HOST>:5432/sapa_db
MONGODB_URI=mongodb://sapa-mongo.sapa.svc.cluster.local:27017
MONGODB_DB=sapa
MONGODB_COLLECTION_ATTENDANCE=attendance_logs
MONGODB_COLLECTION_AUDIT=audit_logs

MQTT_BROKER=sapa-mosquitto.sapa.svc.cluster.local
MQTT_PORT=1883
MQTT_USERNAME=backend
MQTT_PASSWORD=<password kuat>
MQTT_TOPIC_GATE=sapa/gate
MQTT_TOPIC_ATTENDANCE=sapa/attendance
MQTT_TOPIC_DEVICE_STATUS=sapa/device/status
MQTT_TOPIC_DEVICE_HEARTBEAT=sapa/device/heartbeat
MQTT_TOPIC_PIR=sapa/pir
IOT_OFFLINE_AFTER_SECONDS=15

EDGE_INGEST_KEY=<random>
SECRET_KEY=<random panjang>
'''
```

Lalu unit systemd (lihat contoh sebelumnya). Restart:

```
'''bash
sudo systemctl daemon-reload
sudo systemctl enable --now sapa-backend
sudo journalctl -u sapa-backend -f
'''
```

6) Build & Deploy Frontend (Nginx)

```
'''bash
```

```
cd /opt/sapa/frontend
npm ci
npm run build
sudo mkdir -p /var/www/sapa
sudo rsync -a --delete /opt/sapa/frontend/dist/ /var/www/sapa/
'''
```

'/etc/nginx/sites-available/sapa' (sudah benar dari versi sebelumnya): SPA fallback 'try_files \$uri /index.html', reverse proxy '/api/' ? '127.0.0.1:8080'.

```
'''bash
sudo ln -sf /etc/nginx/sites-available/sapa /etc/nginx/sites-enabled/sapa
sudo nginx -t && sudo systemctl reload nginx
sudo certbot --nginx -d sapa.example.com
'''
```

7) Edge Server ? AI Face Recognition di edge

Edge server adalah Python service yang **berjalan di mesin lokal dekat gate** (PC, mini-PC, RPi 4/5, Jetson). Kode-nya ada di folder 'edge_server/'.

7.1 Bagaimana path '/edge' di VPS terhubung ke model AI di edge

- Saat halaman '/edge' di buka di **edge device** (misal PC dekat gate), browser meminta izin kamera, lalu setiap ~350 ms mengirim frame JPEG ke 'POST /api/edge/frame'. Frame ini langsung dilihat di Dashboard (live preview).
- Selain browser-publisher itu, **Edge Server (Python)** menjalankan model AI lokal:
 1. Tarik daftar 'GET /api/employees/' dari VPS.
 2. Tarik gambar wajah 'GET /api/uploads/faces/<id>.jpg'.
 3. Hitung embedding wajah lokal ? simpan di 'embeddings.npz'.
 4. Untuk tiap frame kamera, cocokkan embedding live dengan cache.
 5. Bila match valid ? 'POST /api/edge/face-match {is_valid:true, employee_id, confidence}'.
 6. Backend mencatat log presensi di MongoDB **dan** publish 'sapa/gate {action:"open", employee_id}' lewat MQTT ? ESP32 buka servo + bunyi buzzer pendek.
 7. Bila wajah terdeteksi tapi tak match ? 'is_valid:false' ? backend publish 'sapa/gate {action:"invalid"}' ? ESP32 bunyikan buzzer ganda, servo tetap tutup.

> Model AI tetap di edge (privasi + latensi). VPS hanya menyimpan gambar referensi & menerima keputusan.

7.2 Install di edge

```
'''bash
git clone <REPO_URL> /opt/sapa
cd /opt/sapa/edge_server
python3.11 -m venv venv
./venv/bin/pip install -r requirements.txt
cp .env.example .env
nano .env # isi API base, edge key, MQTT broker (LAN/VPN)
'''
```

Linux dependencies untuk 'face_recognition':

```
'''bash
sudo apt install -y cmake libopenblas-dev liblapack-dev libgl1
'''
```

7.3 Jalankan

```
'''bash
./venv/bin/python -m edge_server.main
'''
```

Service akan:

- konek MQTT broker (publish heartbeat ke 'sapa/device/heartbeat' setiap 5s ? itu yang membuat tombol Open/Close di Dashboard menjadi **enabled** dan badge **Gate ONLINE** menyala);
- sync embedding setiap 60 detik;
- push frame 2 fps ke '/api/edge/frame';
- match wajah lalu push hasil ke '/api/edge/face-match'.

8) ESP32 (servo + buzzer + PIR) ? 'esp32/sapa_gate.ino'

- Servo di GPIO 13, buzzer GPIO 14, PIR GPIO 27.
- Ubah 'WIFI_SSID/PASS' dan 'MQTT_HOST/USER/PASS' lalu flash via Arduino IDE.
- Subscribe 'sapa/gate':
 - '{"action":"open"}' ? servo ke OPEN_ANGLE, beep 1x, hold sampai PIR mendeteksi pergerakan, lalu close (mirip KAI Access).
 - '{"action":"close"}' ? servo ke CLOSED_ANGLE.
 - '{"action":"invalid"}' ? buzzer beep 2x, servo tetap tutup.
- Publish 'sapa/device/heartbeat' setiap 5s ? backend tahu device online.
- Publish 'sapa/pir' saat PIR HIGH ? backend juga publish 'close' ke gate (server-side fail-safe).

8.1 ESP32 LAN ke broker

Gunakan 'mosquitto-nodeport.yaml' (NodePort 31883), UFW allow '192.168.0.0/24' ke port itu, MQTT_HOST = IP VPS LAN.

8.2 ESP32 beda jaringan

Pasang gateway WireGuard / mini router yang join VPN, ESP32 connect ke gateway via LAN; gateway forward ke broker via tunnel.

9) Login Audit (Username, Status, Timestamp) di VPS

- Endpoint 'POST /login' sekarang menulis dokumen ke MongoDB collection 'audit_logs':

```
'''json
{ "username": "manager", "status": "success", "timestamp": "2026-05-23T..." , "event_type":
"login_success", ... }
{ "username": "admin", "status": "failed", "timestamp": "...", "event_type":
"login_failed", ... }
'''
```
- Endpoint 'GET /audit/logins' (manager only) mengambil 200 entry terbaru. Dashboard tab **System ? Login Activity** memuatnya dan auto-refresh tiap 10 detik.

Backup: 'mongodump --db sapa --collection audit_logs' jadwalkan via CronJob k8s atau systemd timer.

10) Realtime log presensi di Dashboard

- Frontend sudah polling 'GET /attendance/' setiap 5 detik ? log update otomatis.
- Backend menulis tiap face match valid (dari edge) ke MongoDB 'attendance_logs' dengan field 'source: edge_ai|edge_mqtt|manual', 'confidence'.
- Tab Logs Dashboard punya filter (date / direction / status / employee) + ekspor CSV.

11) Verifikasi end-to-end

1. **Login audit**: login dengan kredensial salah lalu benar; cek di tab System ? Login Activity.
2. **IoT status**: matikan ESP32 selama >15s. Badge berubah ke **Gate OFFLINE** dan tombol Open/Close ter-disable. Hidupkan kembali ? ONLINE.
3. **Manual gate**: pencet Open/Close (saat ONLINE). ESP32 servo bergerak, buzzer beep.
4. **Edge live preview**: buka '/edge' di edge device, dashboard menampilkan frame.
5. **Face match**: jalankan edge server. Arahkan wajah employee terdaftar. Servo membuka, log baru muncul realtime di dashboard. Wajah asing ? buzzer ganda, log tidak bertambah.
6. **PIR auto-close**: setelah gate dibuka, lewatkan tangan di depan PIR ? servo menutup dan tab System menampilkan 'last_action: auto_close_pir'.
7. **View Faces**: di tab Employees, klik **View Faces** ? modal menampilkan foto referensi '/uploads/faces/<id>.jpg'.

12) Hardening

- Set 'EDGE_INGEST_KEY' panjang dan rahasiakan; edge server kirimkan via field 'edge_key' / header 'X-EDGE-KEY'.
- Mosquitto: 'allow_anonymous false', 'password_file', 'acl_file' untuk membatasi user 'backend' (publish 'sapa/gate', subscribe sisanya) dan 'esp32' (subscribe 'sapa/gate', publish 'sapa/device/heartbeat', 'sapa/pir').
- WireGuard untuk ESP32 jika berbeda jaringan.
- HTTPS wajib agar '/edge' (browser) bisa minta izin kamera ('getUserMedia').
- Backup MongoDB & PostgreSQL terjadwal.

13) Troubleshooting cepat

```
| Gejala | Periksa |  
|-----|-----|  
| Tombol Open/Close ter-disable | Edge/ESP32 tidak publish heartbeat. 'sudo journalctl -u  
sapa-backend -f' lihat apakah on_message dapat MQTT. |  
| Live frame kosong | '/edge' di edge device tidak terbuka, atau backend tidak menerima  
'/edge/frame' (cek Nginx access log + 'EDGE_INGEST_KEY'). |  
| Wajah employee tak dikenali | Edge server belum sync ('./venv/bin/python -m edge_server.main'  
dan tunggu sync log). Threshold terlalu ketat: turunkan 'RECOGNITION_THRESHOLD'. Foto referensi  
tidak ada wajah jelas: re-upload dari Add Employee. |  
| Login audit kosong | MongoDB tidak terjangkau. 'kubectl -n sapa get pods' & 'MONGODB_URI' di  
backend '.env'. |  
| Servo tetap diam saat valid | Cek topic match: 'mosquitto_sub -h <broker> -t sapa/gate -v'  
saat scan wajah. |
```